Project Delivery:AZ-500 Service Endpoints and Securing Storage

FuturinCLOUD Limited
Region: West US

Executive Summary

This lab demonstrated how to secure an Azure File Share by combining Virtual Network Service Endpoints and Network Security Group (NSG) rules to enforce private connectivity and strict subnet-level isolation.

The solution ensures that:

- All traffic to the storage account remains on the Azure backbone (never traverses the public internet)
- Only VMs in the designated private subnet can reach the file share
- VMs in any other subnet (including public ones) are explicitly denied access

The lab was completed successfully with clear validation of both allowed and denied access scenarios.

Project Deliverables

- Resource group: futuricloud-lab5-rg
- Virtual Network: futuricloud-vnet (10.0.0.0/16)
- Subnets:
  - private-subnet (10.0.1.0/24) – configured with Microsoft.Storage service endpoint
  - public-subnet (10.0.2.0/24) – no endpoint
- NSGs:
  - futuricloud-private-nsg – allows SMB (port 445) to Storage service tag
  - futuricloud-public-nsg – allows RDP (port 3389) for testing
- Storage Account: futuricloudstorage001
- File Share: securefileshare
- Test VMs:
  - vm-private (in allowed subnet)
  - vm-public (in denied subnet)

Key Outcomes Achieved

- Private connectivity enforced: File share traffic uses Azure Private Link/service endpoint (no public internet exposure)
- Subnet isolation achieved: Only private-subnet VMs can mount the file share
- Access validation:
  - vm-private → successfully mounts \futurincloudstorage001.file.core.windows.net\securefileshare
  - vm-public → connection fails (timeout/access denied)
- Public exposure eliminated: Storage account public access disabled or restricted via endpoint
- Demonstrated defense-in-depth at the network layer with minimal configuration overhead

Task 1: Created a virtual network

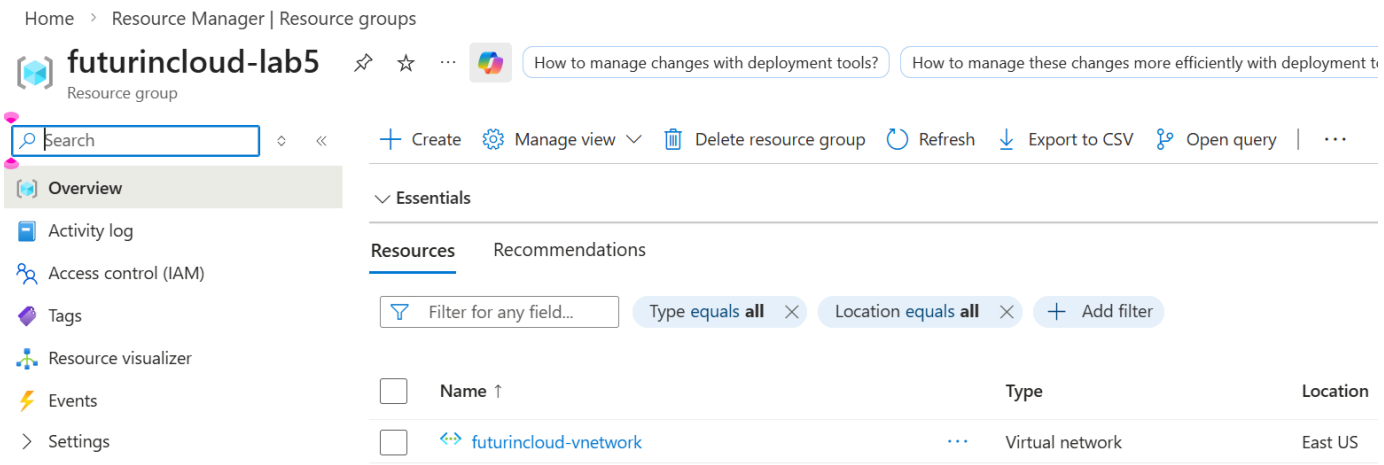


Figure 1: Resource group overview showing futuricloud-virtual network

Task 2: Added a subnet to the virtual network and configure a storage endpoint

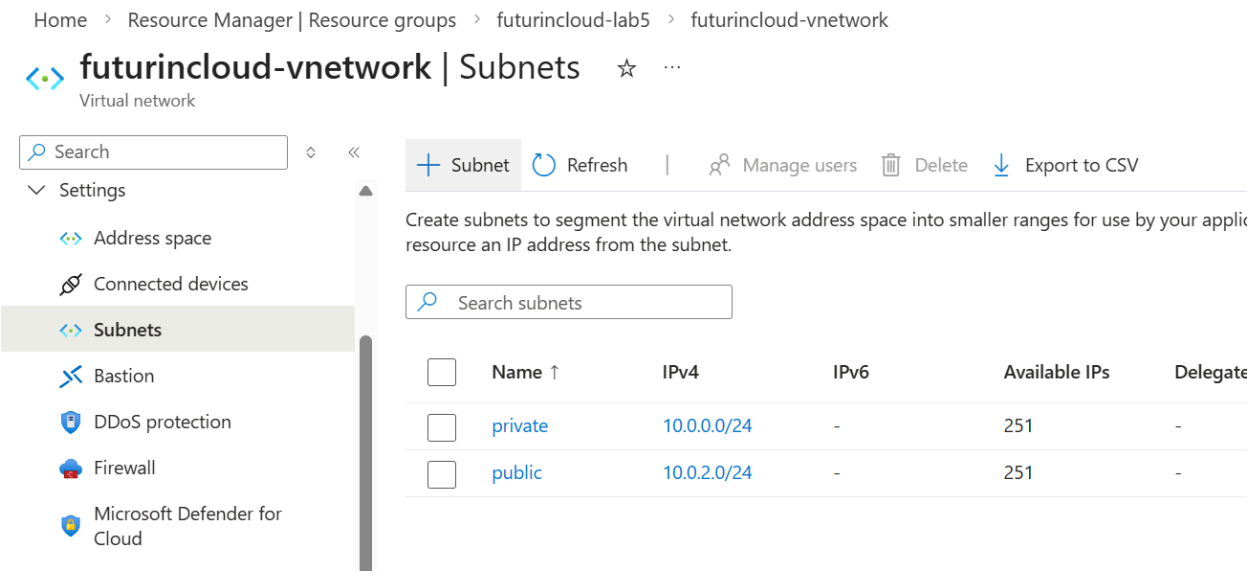


Figure 2: Added two subnets to the Virtual network

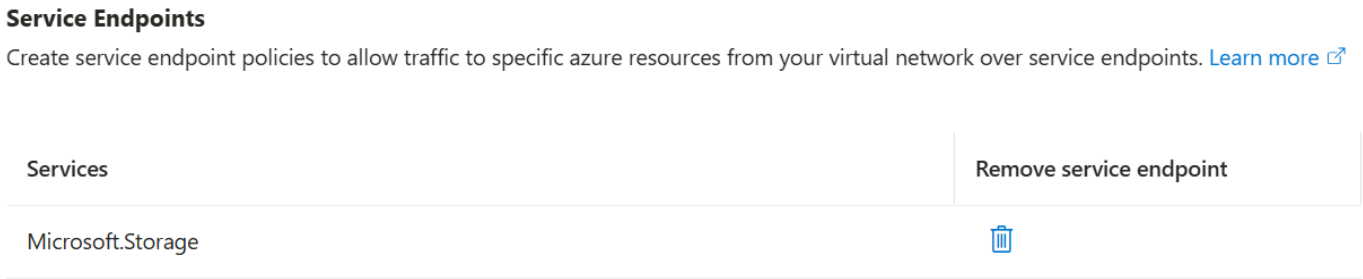


Figure 3: private-subnet settings showing Microsoft.Storage service endpoint enabled

Task 3: Configured a network security group to restrict access to the subnet

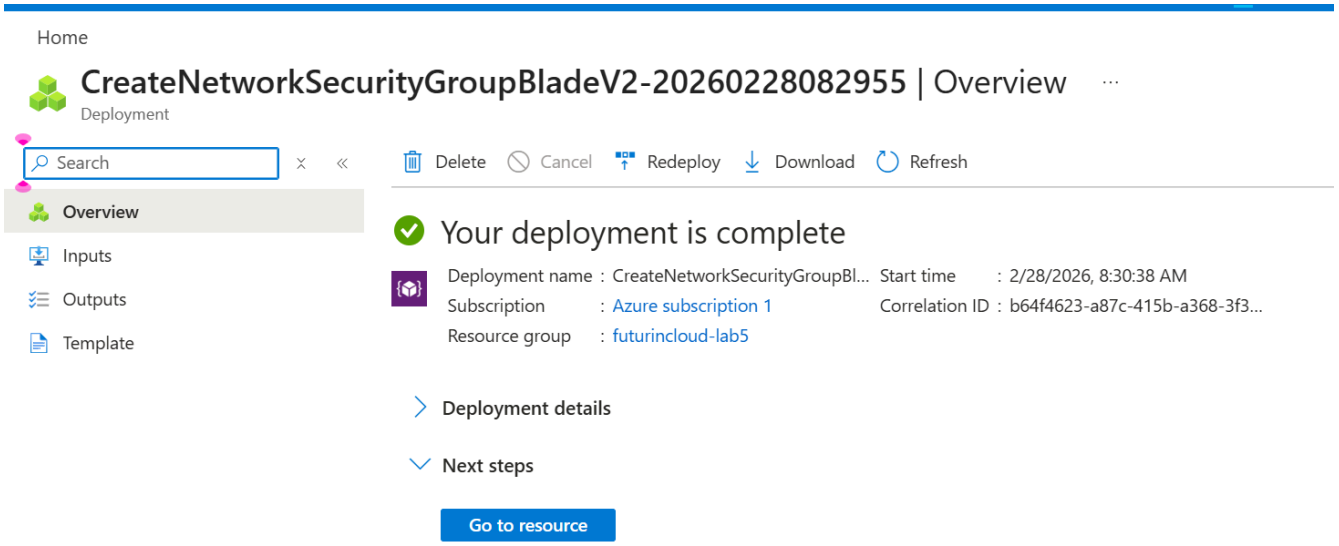


Figure 4: successfully deployed a network security group

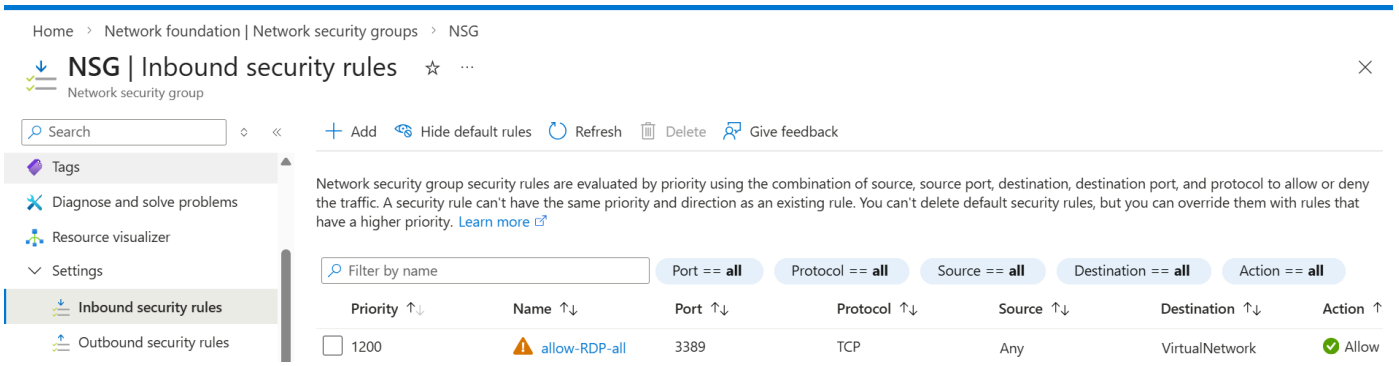


Figure 5: : futuricloud-private-nsg inbound rules showing Allow-RDP-all

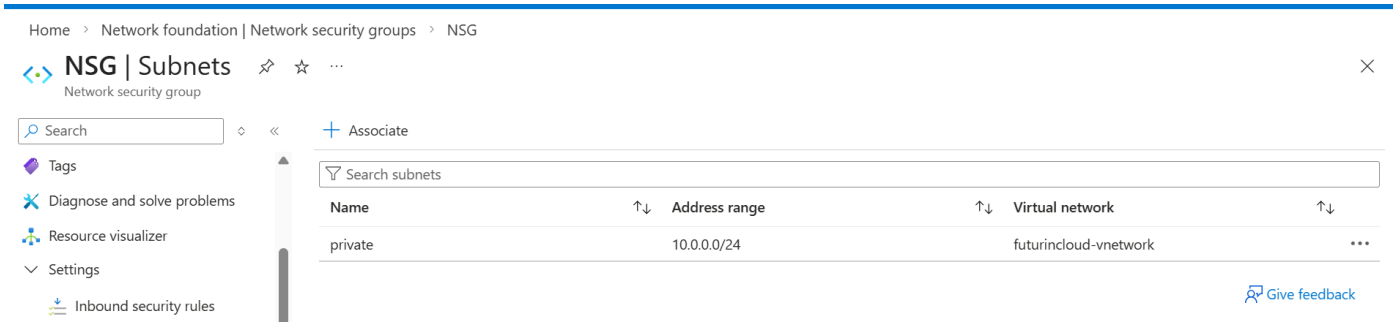


Figure 6: Associated the NSG rule to the private Subnet

Task 4: Configured a network security group to allow rdp on the public subnet

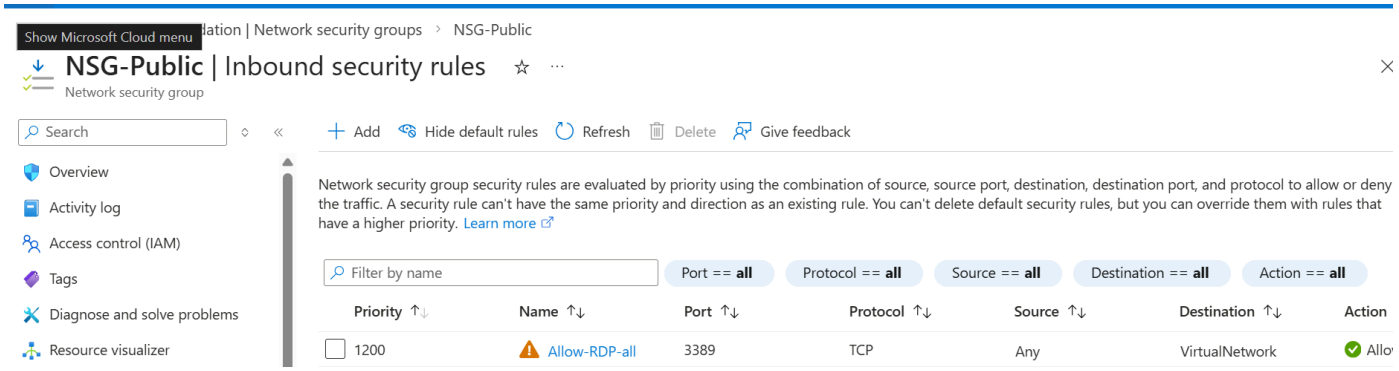


Figure 7: Created futuricloud-public-nsg with inbound rule allowing TCP 3389 (RDP) from Any

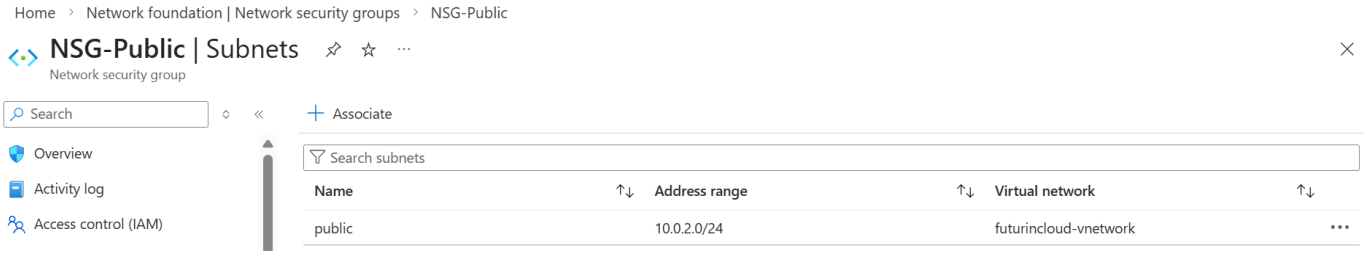


Figure 8: Associated to public-subnet.

Task 5: Created a storage account with a file share

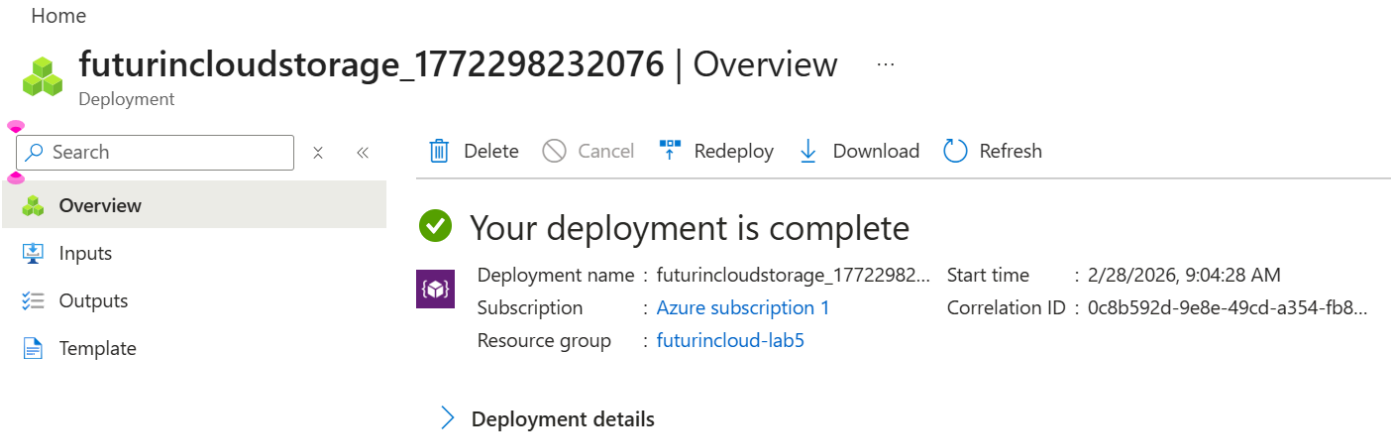


Figure 9: Deployed a storage account

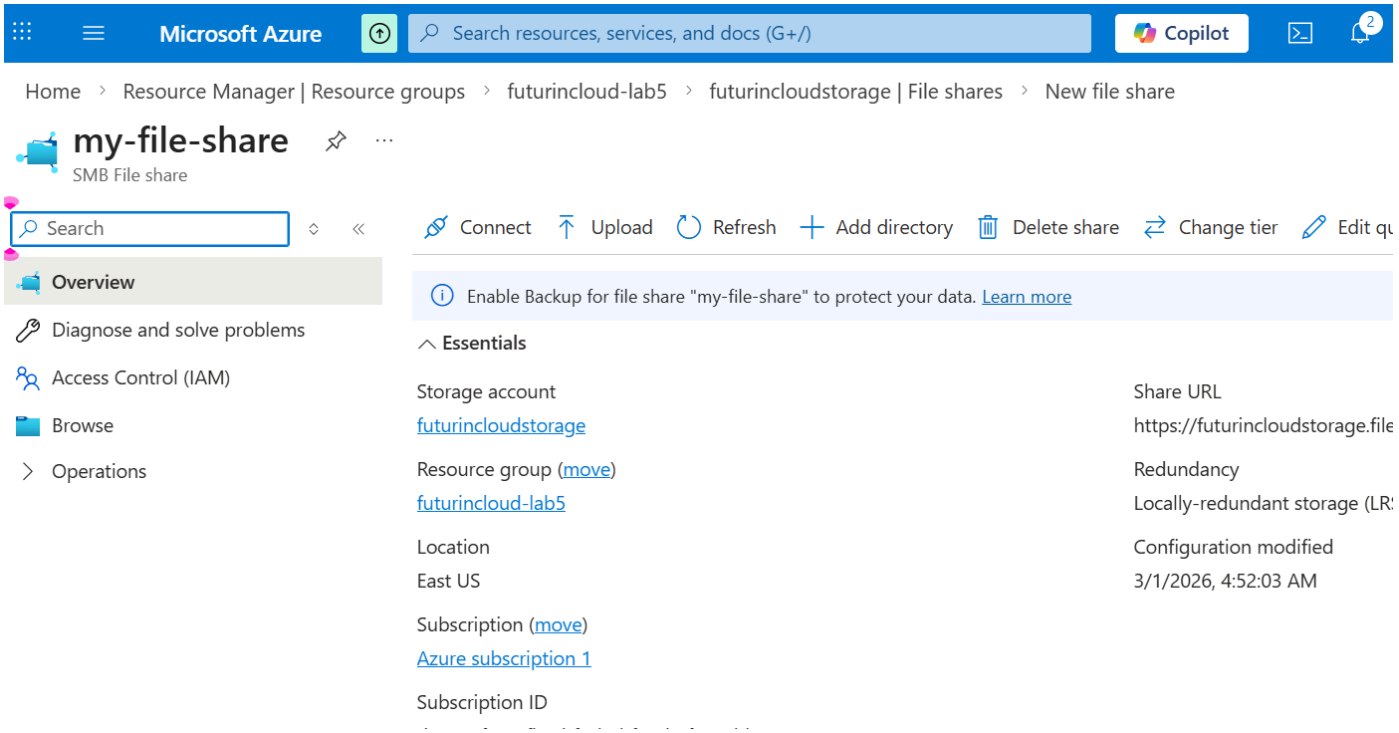
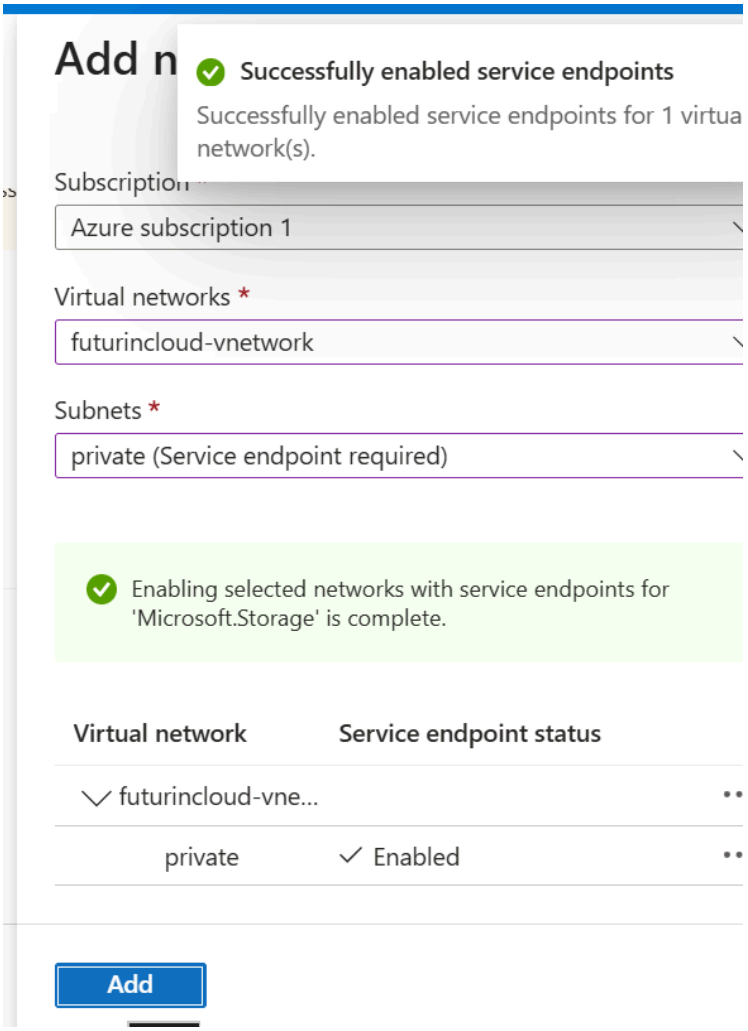


Figure 10: Storage account overview, file share securefileshare visible



Task 6: Deployed virtual machines into the designated subnets

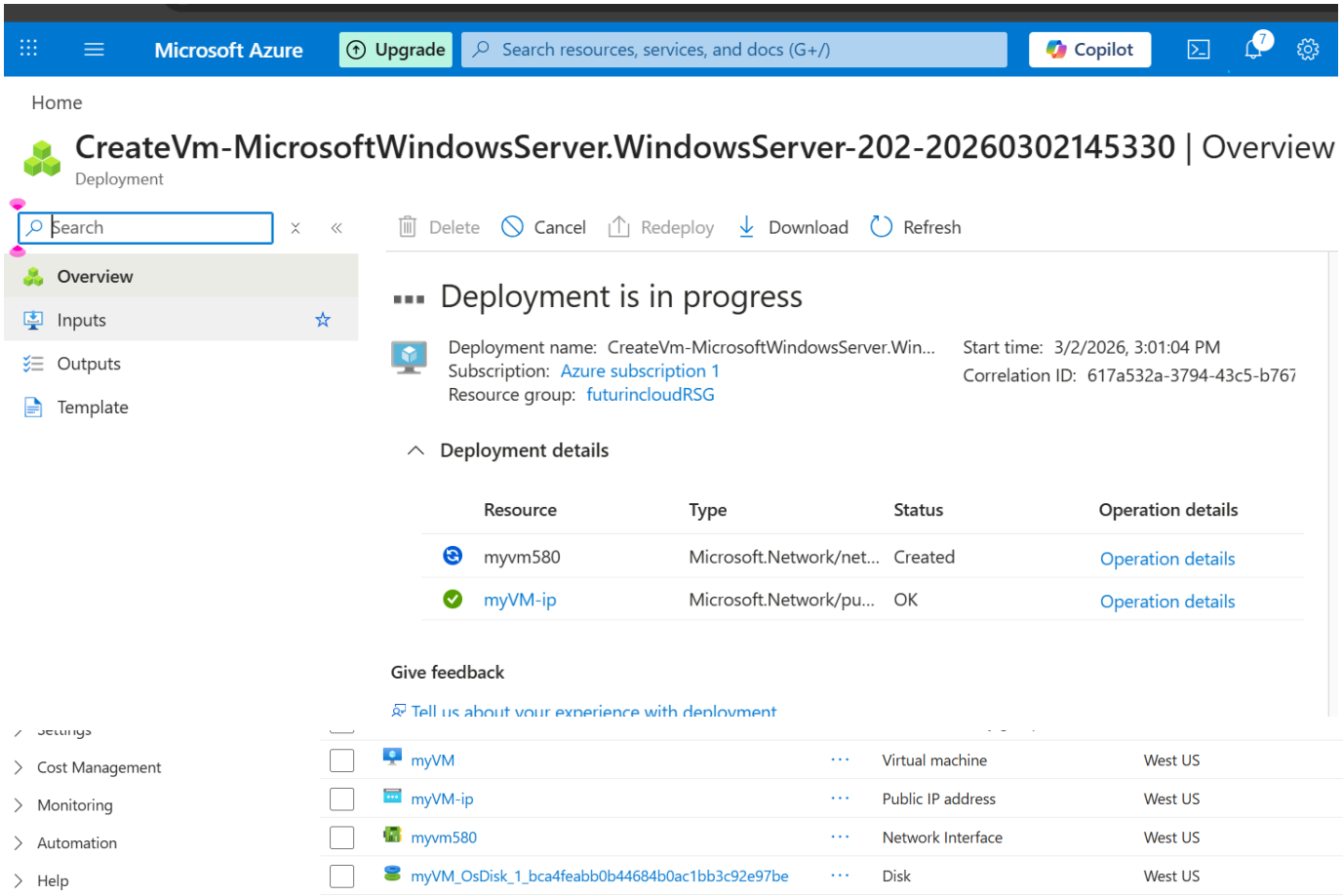
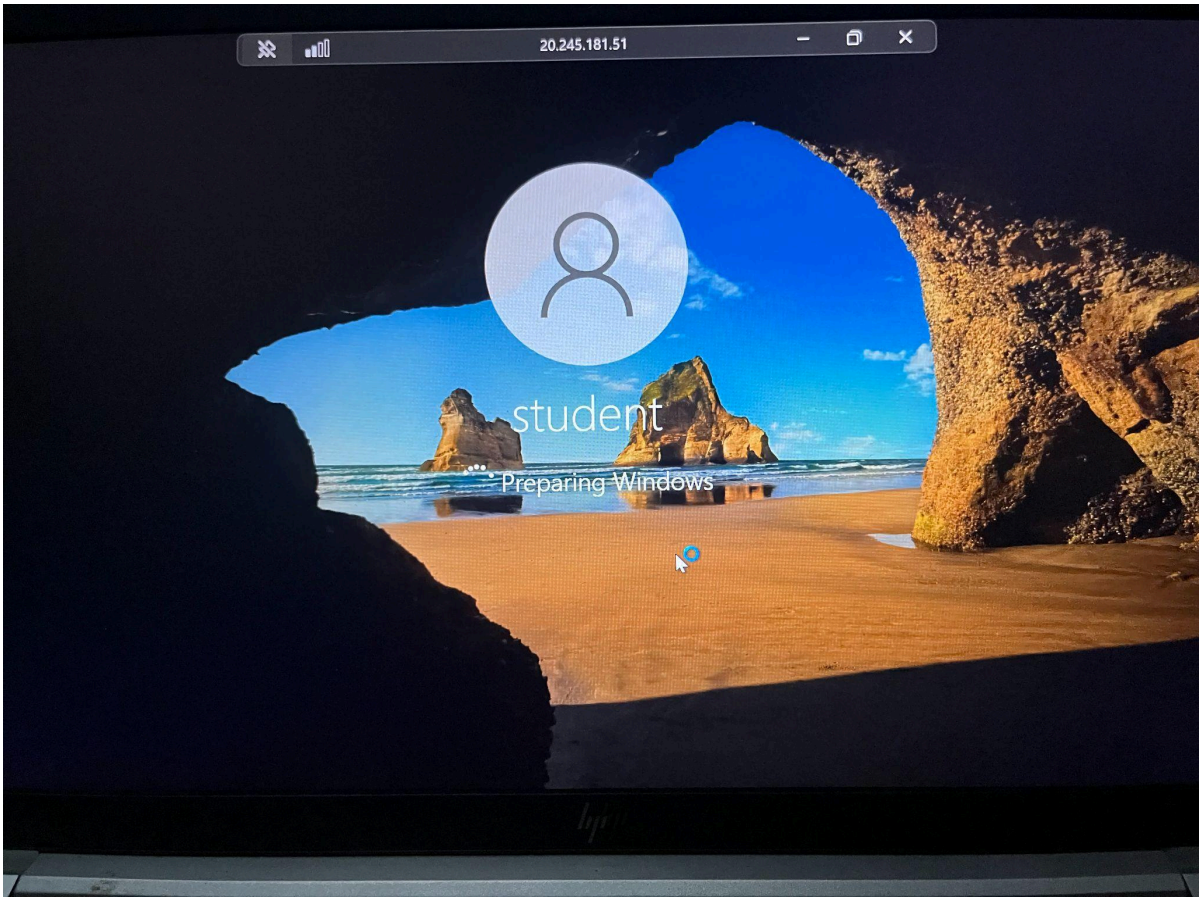
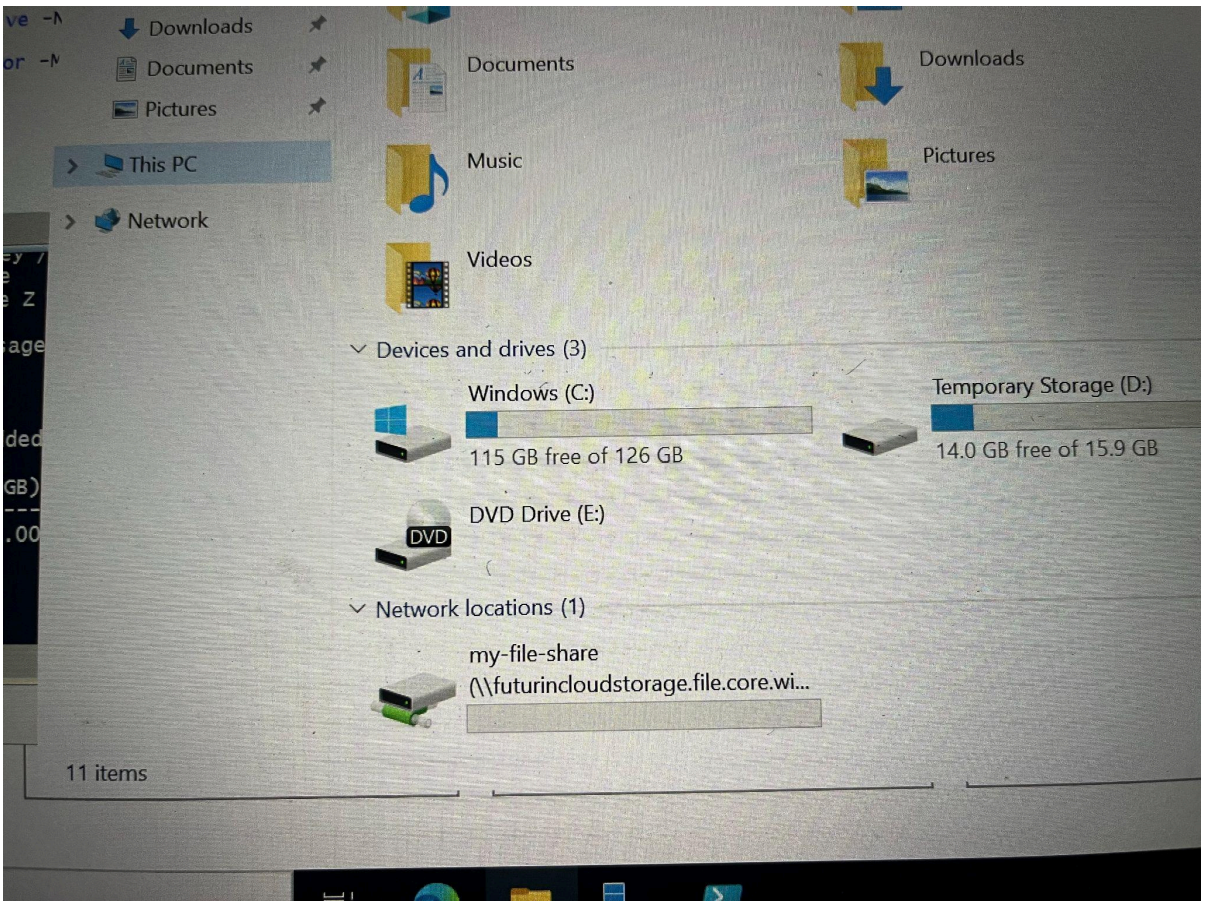
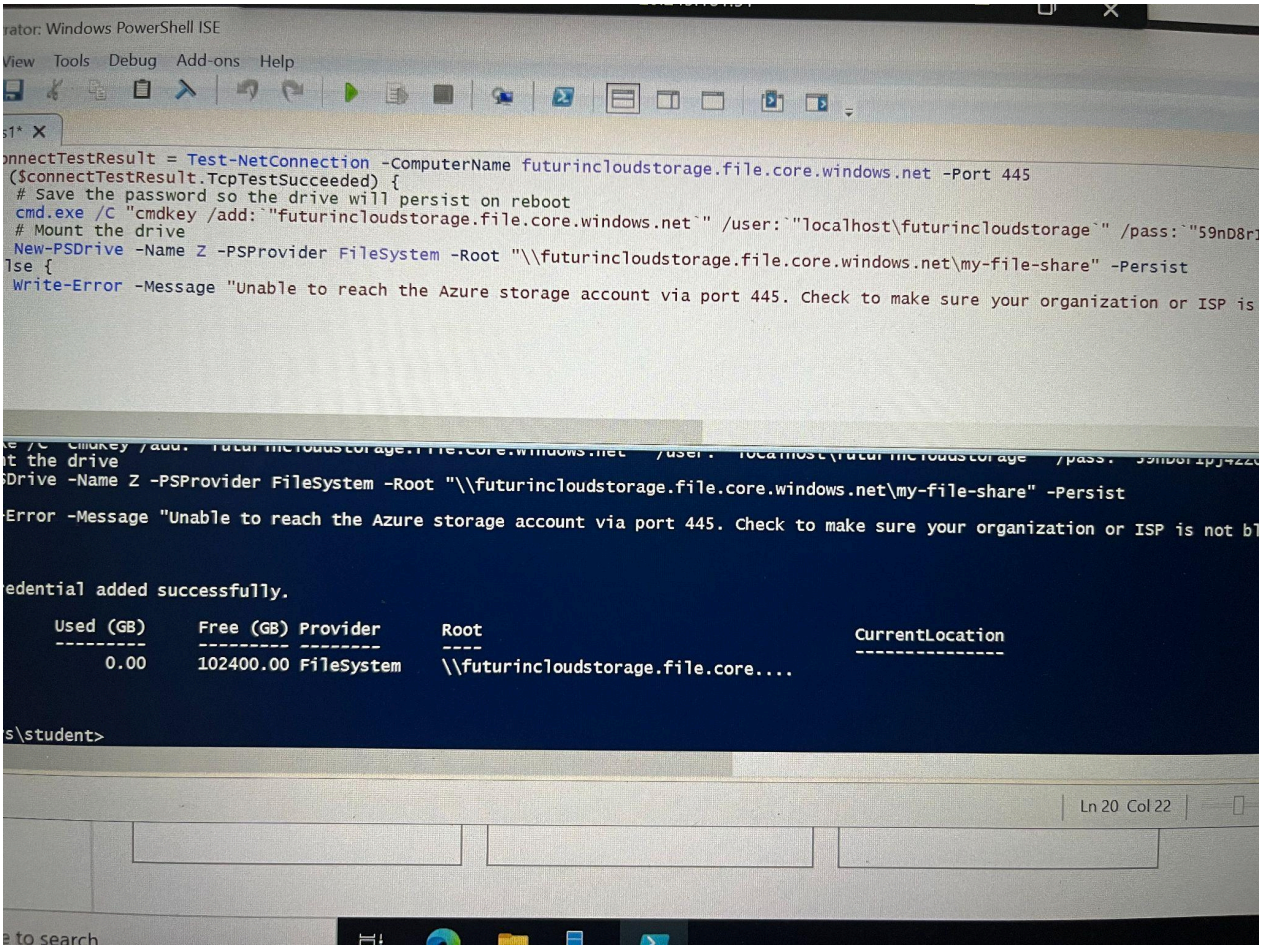


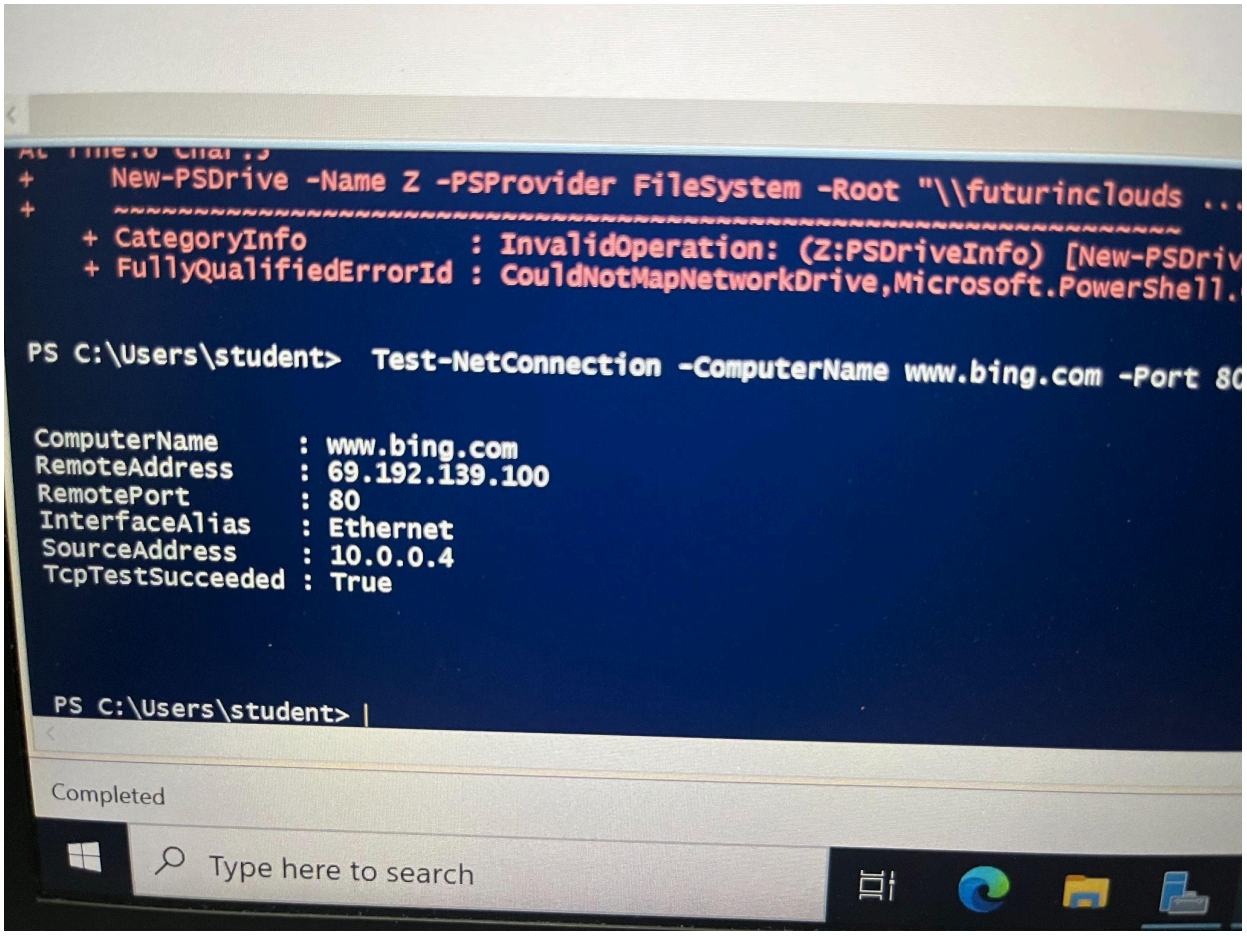
Figure 11&12: Deployed vm-private into private-subnet and vm-public into public-subnet.

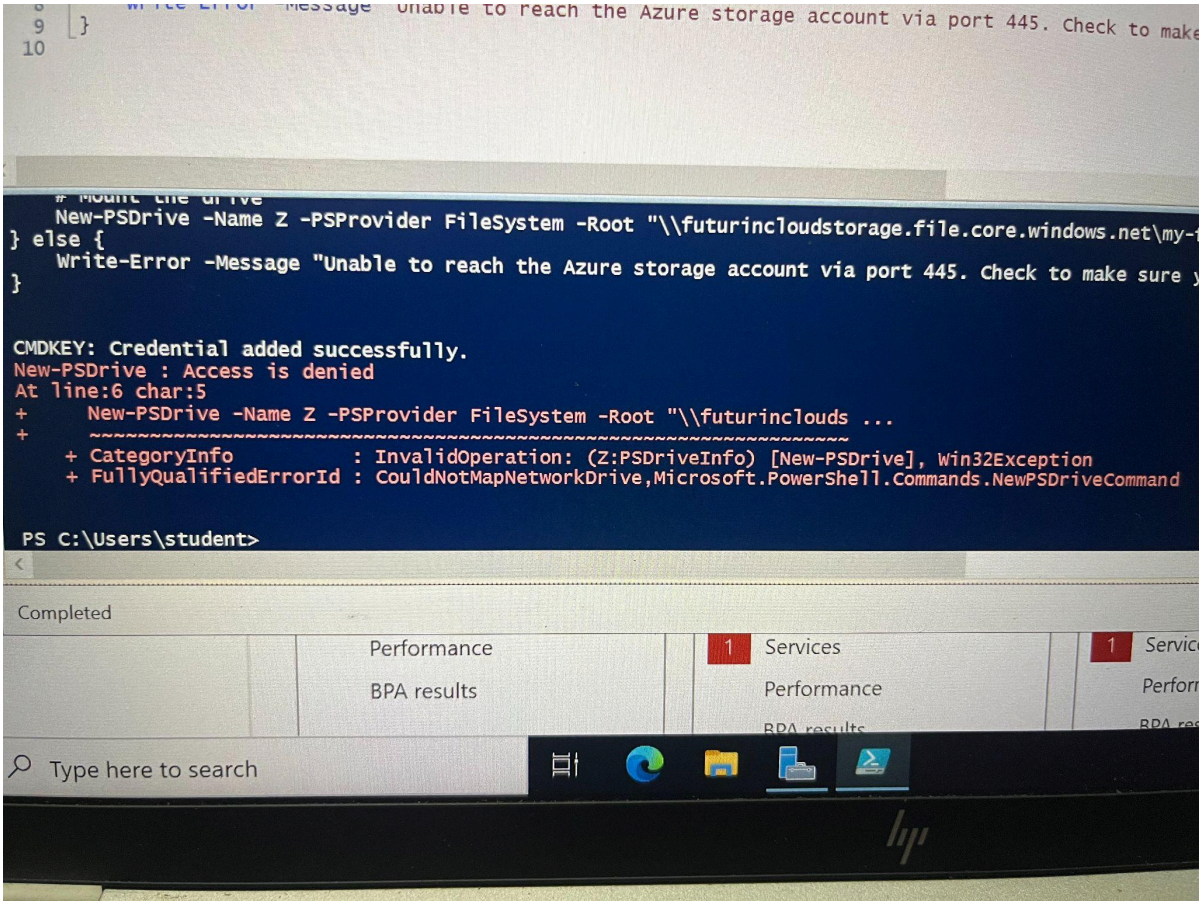
Task 7: Tested the storage connection from the private subnet to confirm that access is allowed


20.245.181.51
student
Preparing Windows


Downloads
Documents
Pictures
This PC
Network
Documents
Downloads
Music
Pictures
Videos
Devices and drives (3)
Windows (C:)
115 GB free of 126 GB
Temporary Storage (D:)
14.0 GB free of 15.9 GB
DVD Drive (E:)
DVD
Network locations (1)
my-file-share
(\\futurincloudstorage.file.core.wi...
11 items




+ New-PSDrive -Name Z -PSProvider FileSystem -Root "\\futurinclouds ...
+ ~~~


Write-Error -Message "Unable to reach the Azure storage account via port 445. Check to make
}
# Mount the drive
New-PSDrive -Name Z -PSProvider FileSystem -Root "\\futurincloudstorage.file.core.windows.net\my-
} else {
Write-Error -Message "Unable to reach the Azure storage account via port 445. Check to make sure
}
CMDKEY: Credential added successfully.
New-PSDrive : Access is denied
At line:6 char:5
+ New-PSDrive -Name Z -PSProvider FileSystem -Root "\\futurinclouds ...
+ ~~~

- Service endpoint propagation delay Waited 1–2 minutes after enabling, refreshed VM networking page, re-tested
- NSG rule not applying immediately → Checked effective security rules on VM NIC, waited for propagation

Conclusion

This Lab successfully demonstrated how to isolate an Azure File Share using service endpoints and NSG rules to enforce private connectivity and subnet-level access control.

The solution ensures:

- No public internet exposure for storage traffic
- Only authorized subnets can reach the file share
- Clear validation of allowed vs. denied access scenarios